

Lab G: Stream Discharge Prelab Reading

The amount of water in a stream channel is related to the size of the area it drains and the average precipitation over that area. The area drained by a stream and its smaller streams called tributaries is called a drainage basin. Drainage basins are separated by regional topographic high ground known as a drainage divide. For example, the Mississippi River is separated from streams that drain to the west by a drainage divide represented by the Rocky Mountains. Streams west of the Rockies flow to the Pacific Ocean or Gulf of California. Streams in the eastern U.S. flow to the Gulf of Mexico or the Atlantic Ocean.

The size of the drainage basin typically increases as the size of the stream increases. The larger Mississippi River is fed by other large rivers that have their own drainage basins within the boundaries of the Mississippi River drainage basin. The Ohio and Missouri and other rivers are tributary streams to the Mississippi River. Streams simply carry water from higher elevations to lower elevations. The slope of a stream is known as the stream gradient and is the change in elevation of the stream over a horizontal distance. Stream gradients typically range from 10s of meters per kilometer (~50-100 feet per mile) near the source, to a few centimeters per kilometer (inches per mile) near the river's mouth.



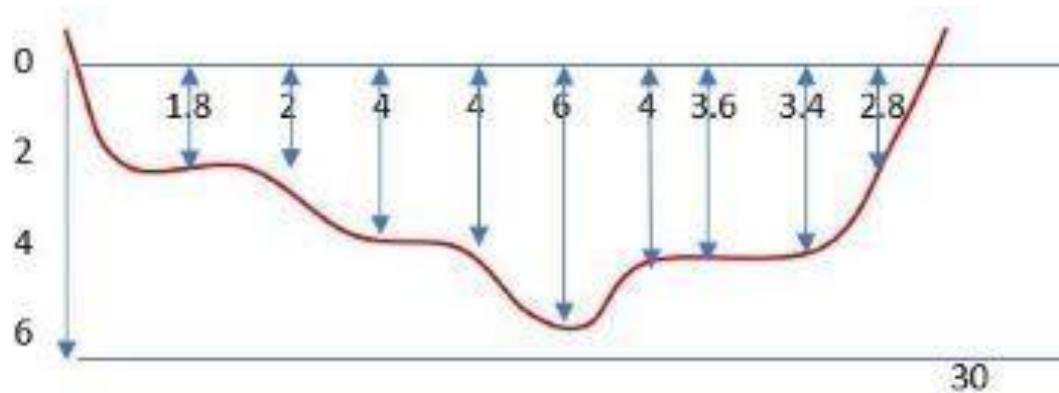
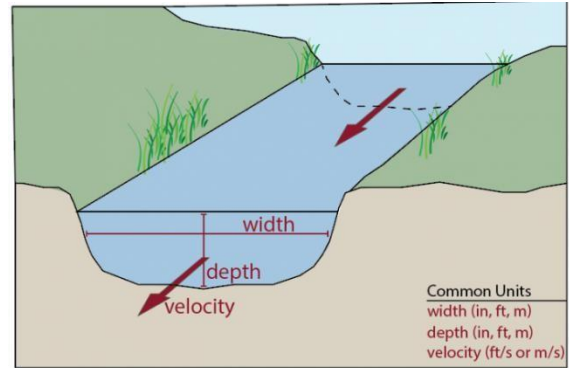
Stream Characteristics

The velocity of flow in any stream channel is determined by timing how long it takes for an object to travel a set distance. For example, how far a rubber duck will float in a second (meters per second). Velocity is seldom uniform across a stream channel so you will have to take multiple measurements to determine the average velocity for a chosen stream segment. The amount of water flowing in a stream can be measured as stream discharge. Discharge is the volume of water that flows past a given point in a given time. The volume is calculated by multiplying the cross-sectional area of the stream channel by the velocity of flow.

- Cross sectional area = depth x width.

If the stream channel was a simple rectangle, this would be an easy calculation. However, the channel is more likely to be irregular so you will have to measure depth at several locations across the stream to obtain an average the cross-sectional area.

In the image below, the person took nine depth readings. To calculate the average depth, they would need to add all the depths together and divide by 9. To calculate cross-sectional area, they would need to multiply the average depth by the width of the stream.



- Average depth: $31.9 \text{ m} / 9 = 3.5 \text{ m}$
- Stream width = 30 m
- Average cross-sectional area = $3.5 \text{ m} \times 30 \text{ m} = 105 \text{ m}^2$

Velocity of the stream can be calculated by taking the distance flowed divided by the time it took. For example, if an object traveled 10 m in 5 seconds, the velocity is 2 m/s.

- Velocity = $10 \text{ m} / 5 \text{ s} = 2 \text{ m/s}$

Discharge is calculated by multiplying cross-sectional by velocity ($D=A \times V$)

If we use the information we calculated above, we get the following for discharge.

- Discharge = $105 \text{ m}^2 \times 2 \text{ m/s} = 210 \text{ m}^3/\text{s}$